



ScholarNest

Spark Azure Databricks

Databricks Spark Certification and beyond

Instructor: Prashant Kumar Pandey



Azure
Databricks



Absolute Beginner to Specialization in Apache Spark and Azure Databricks



Data Lake – Emergence and Use

We know that the distributed computing started with Google finding a solution for their storage requirements using GFS. The open-source community created a similar solution called HDFS that allowed us to form a cluster of computers and use the combined capacity for storing our data. Then we also got the MapReduce framework which allowed us to use the combined computing power of the cluster and use it to process the enormous data volumes that we stored in HDFS.

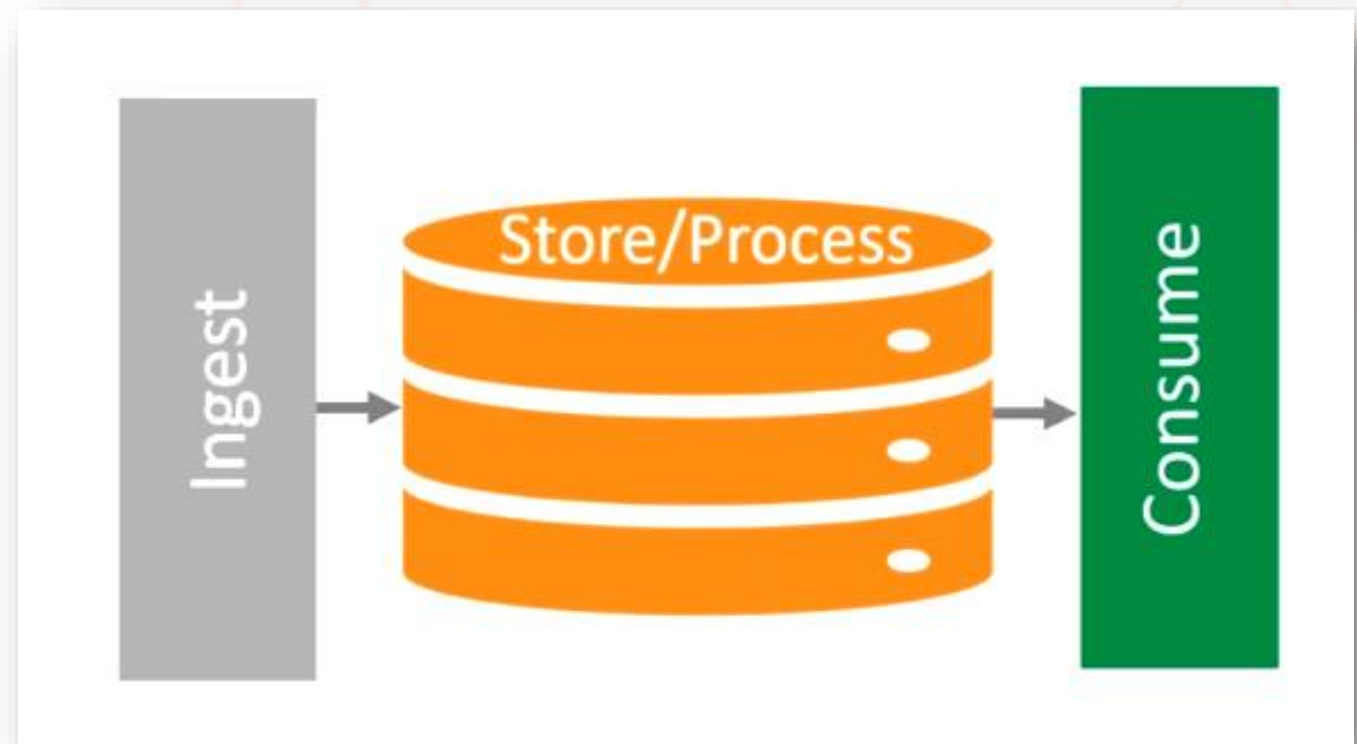


Before the HDFS and Map/Reduce came into existence we had Data Warehouses - Like Teradata and Exadata. We created pipelines to collect data from many OLTP systems and brought them into the Data Warehouse. Then we processed all that data to extract business insights and used it to make the correct business decision.

Hadoop also offered to collect data and process it to extract business insights.

So the advent of HDFS and MR started challenging these Data Warehouses in three critical respects:

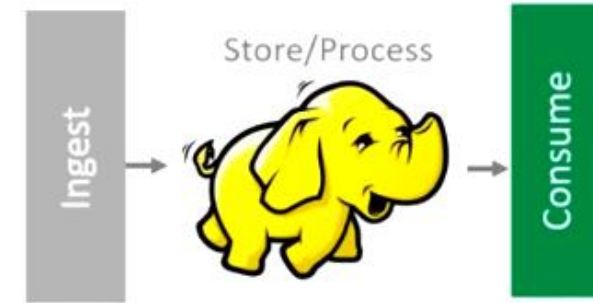
1. Ease of Horizontal Scalability
2. Capital Cost
3. Volume and Variety of Data



Here is a comparison between Data Warehouse and Hadoop offerings.



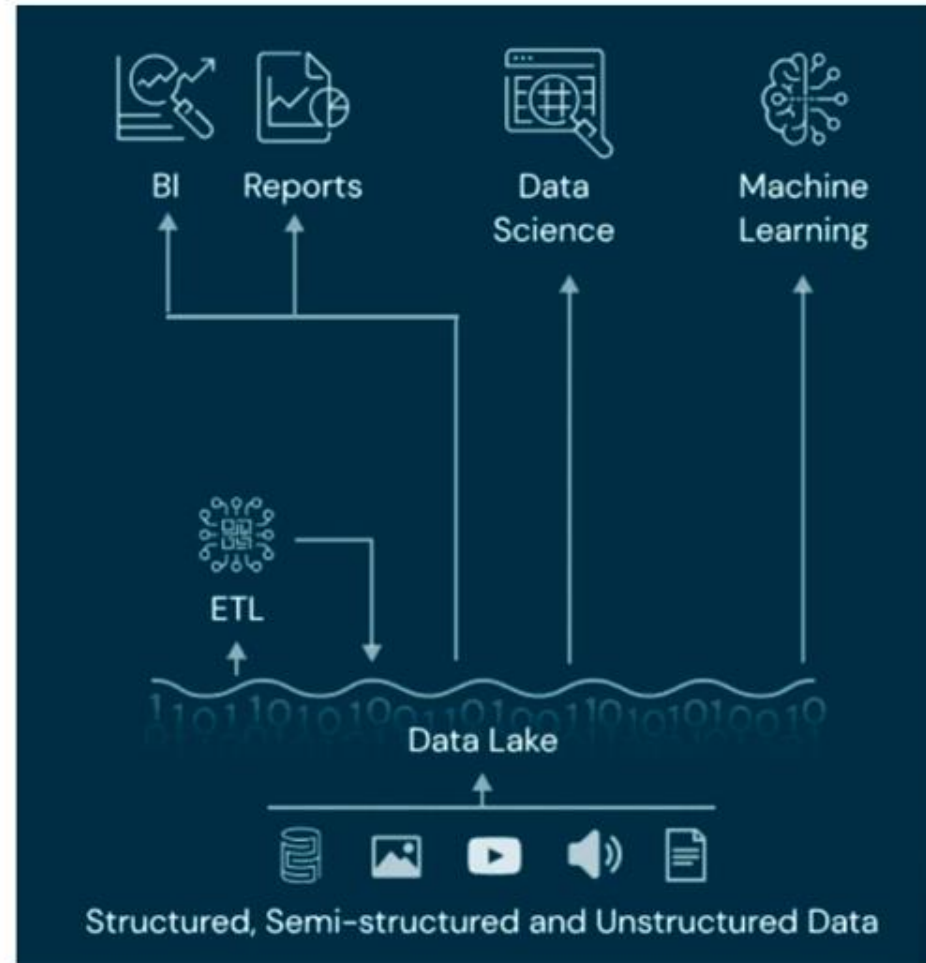
1. Vertical Scaling
2. Large Capital Investment
3. Terabytes to Exabytes
4. Only Structured Data



1. Horizontal Scaling
2. Low Capital Investment
3. Exabytes to Petabytes
4. Structured, Semi Structured and Un Structured Data

Data Warehouses were challenged, and they needed a new name for the Hadoop approach. And it is when the Data Lake came into existence.

Data Lake



Same as Data warehouses, we collected data from different data sources and stored them in HDFS.

Then we used Map Reduce and Spark to process this data and prepare new data models for generating reports and business insights. Spark took over the Map/Reduce and other tools over a period of time, so it is safe to consider that we used Spark to process and prepare data for reporting.

This processed data was also stored in Data Lake storage for business intelligence and reporting.

Most of the popular BI and reporting tools offered a connector to access data from the data lake.

So, Data Lake also allowed us to collect and process huge volumes of semi-structured and unstructured data.

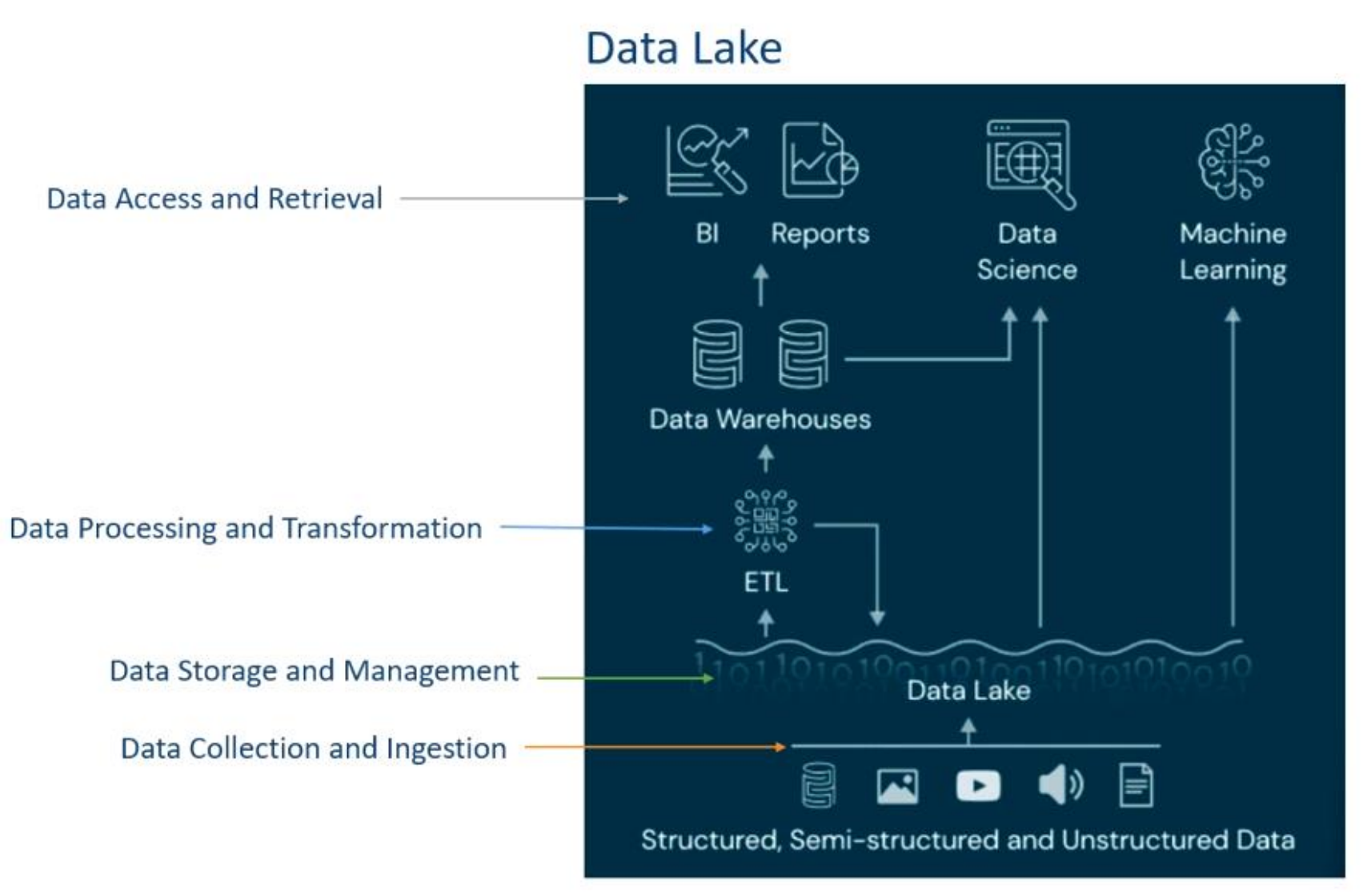
However, the Data Lake technology missed two supercritical features that Data warehouses offered:

1. Transaction and Consistency
2. Reporting Performance

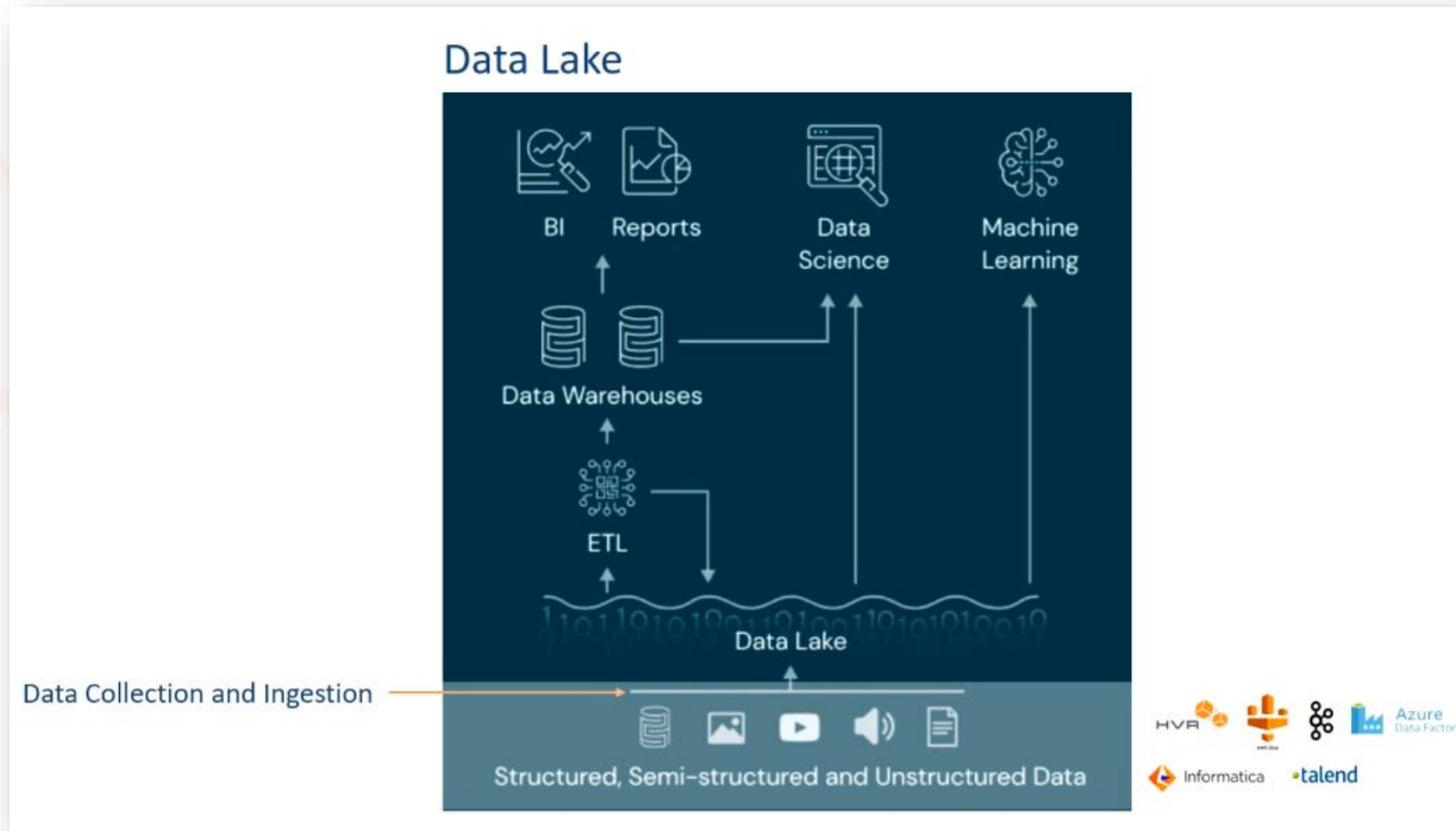
So, we adopted a different architecture for implementing Data Lakes. We collected data in Data lake storage, proceeded it using Apache Spark, and stored the result in a Data Warehouse. And finally, we connected the BI and Reporting with the Data Warehouse.



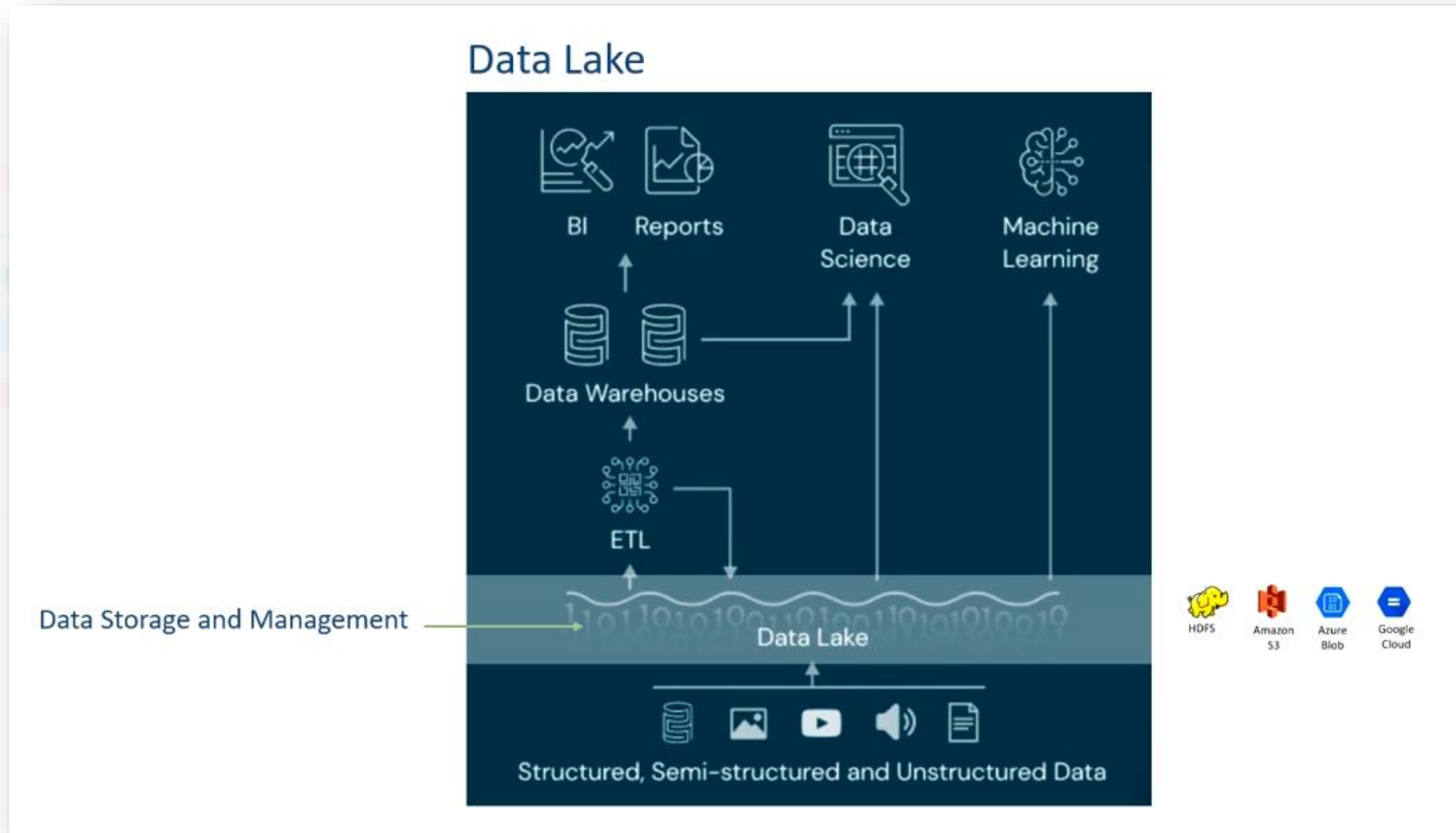
This new architecture also matured as a platform with four key capabilities. And these are nothing but the same problems that Google attacked and solved.



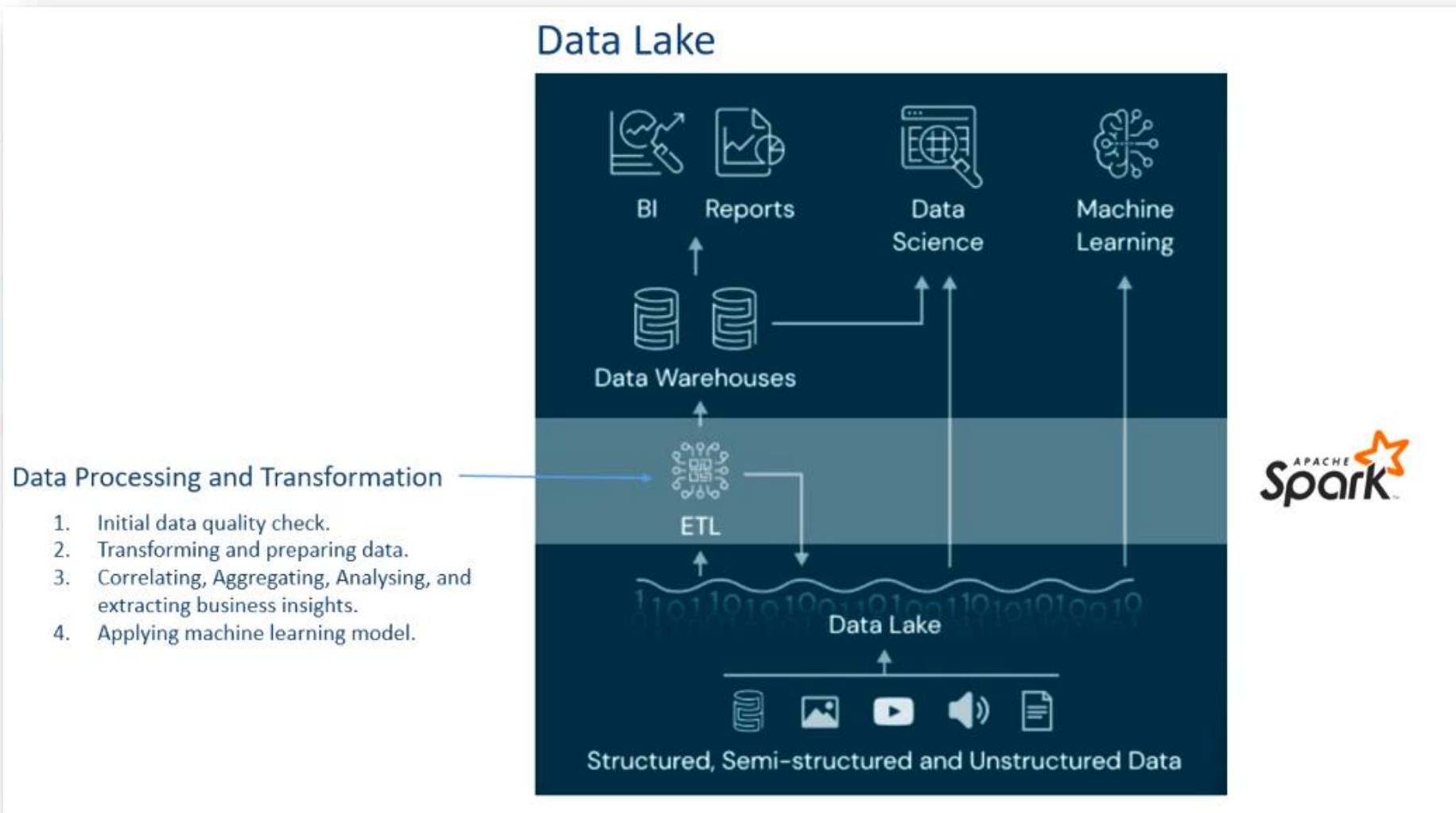
The ingest block of the data lake is all about identifying, implementing, and managing the right tools to bring data from the source systems to the data lake. And we have many vendors competing for a place in this box.



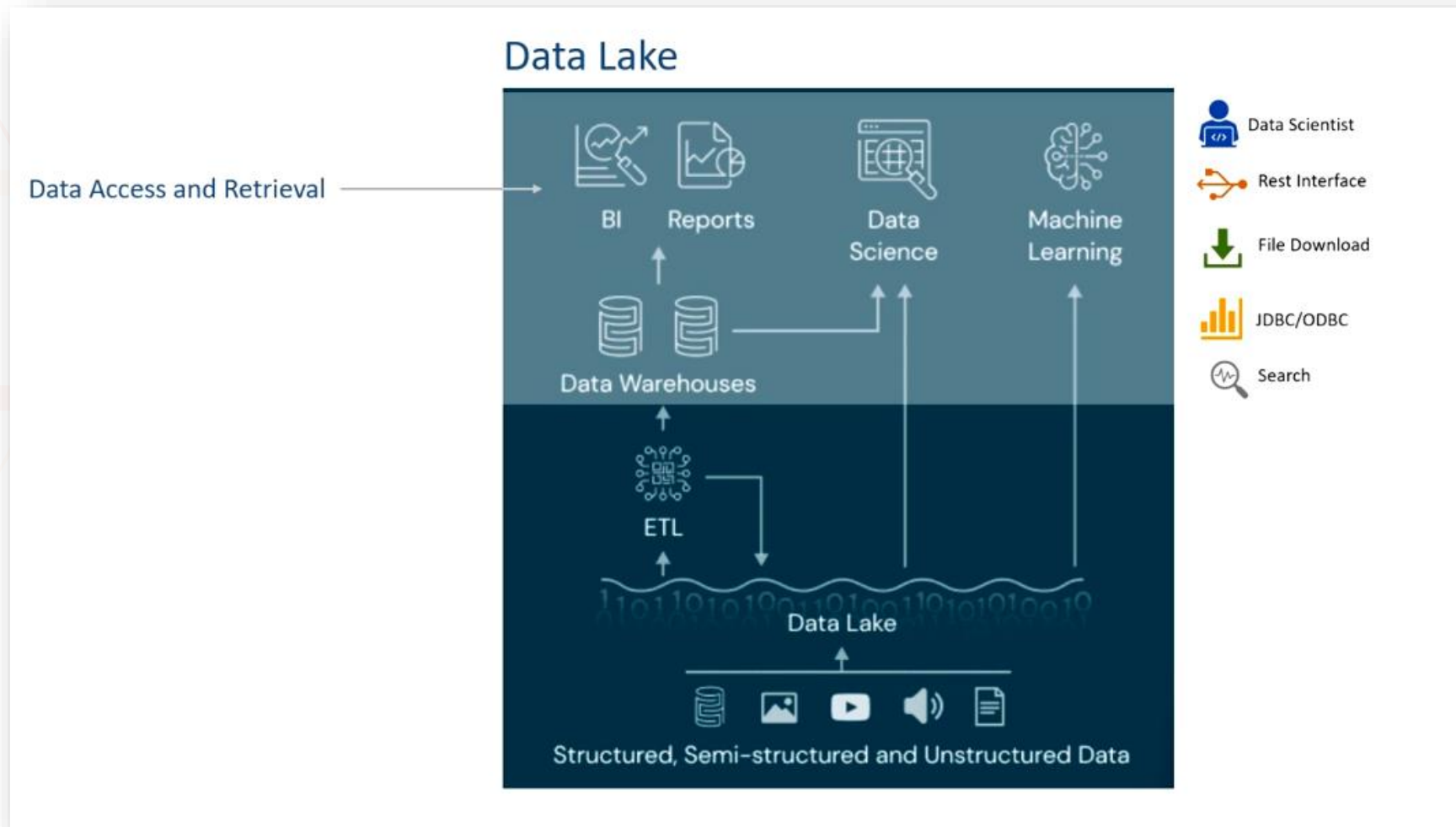
The core of the data lake platform is the storage infrastructure. In today's data lake, this could be an on-premise HDFS or Cloud Object Stores such as Amazon S3, Azure Blob, Azure Data Lake Storage, or Google Cloud Storage. Cloud storage is leading because they offer scalable and high availability access at an extremely low cost in almost no time to procure.



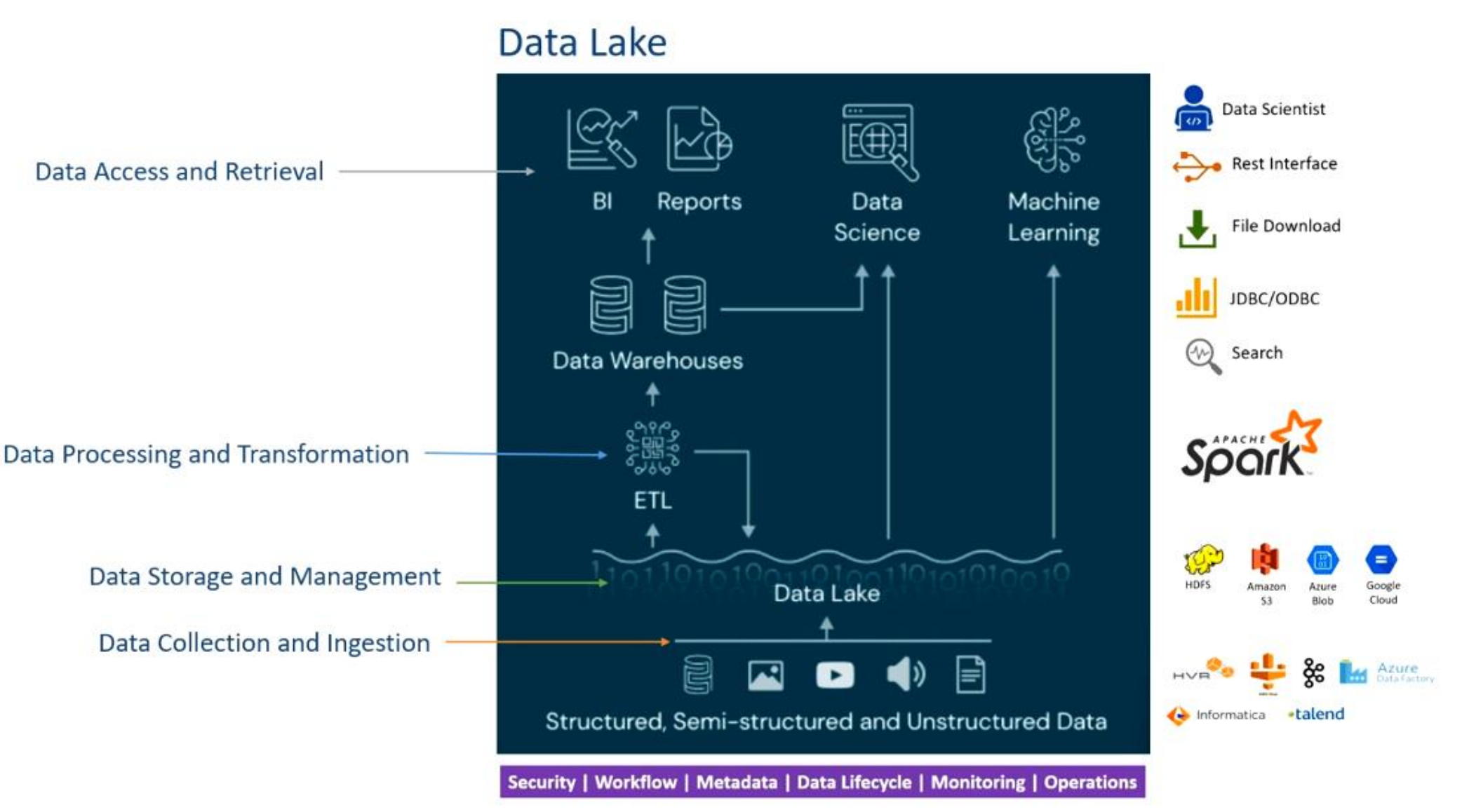
The next layer is the processing layer. This is the place where all the computation is going to happen. When I say computation, it means the following kind of work listed in the image below.



The last and most critical capability of the data lake is to allow you to consume the data from the data lake. You can think of your data lake as a repository of raw and processed data. Now the consumption is all about putting that data for real-life usage, which can be of various types.



This is the complete Data Lake architecture.





Thank You
ScholarNest Technologies Pvt Ltd.
www.scholarnest.com